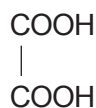


9. (a) Ethanedioic acid is an important industrial chemical.



One method for its manufacture is by the reaction of butan-1-ol, carbon monoxide and oxygen at a pressure of 10 MPa and at a temperature of 100 °C in the liquid phase. The process uses a catalyst of palladium mounted on graphite.

The first stage gives the ester di-1-butyl ethanedioate, which is then hydrolysed to give ethanedioic acid and butan-1-ol.

- (i) Balance the equation for the first stage of this process. [1]



- (ii) Suggest **one** reason why this stage of the process may be seen as relatively uneconomic. [1]

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- (iii) Hydrolysis of the ester gives ethanedioic acid and butan-1-ol.

Explain why ethanedioic acid is soluble in water but di-1-butyl ethanedioate is virtually insoluble in water. Include a diagram in your answer. [3]

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- (b) An older method for manufacturing ethanedioic acid produces insoluble calcium ethanedioate (M_r 128) as an intermediate. This is reacted with aqueous sulfuric acid to produce calcium sulfate and an aqueous solution of ethanedioic acid.



Ethanedioic acid crystallises as its dihydrate.

In a laboratory experiment 11.52 g of calcium ethanedioate was reacted with aqueous sulfuric acid of concentration 2.00 mol dm^{-3} .

- (i) Calculate the minimum volume of this aqueous sulfuric acid needed to just react with all the calcium ethanedioate. [2]

Minimum volume = cm^3

- (ii) After removal of insoluble calcium sulfate, the residue was washed with water. The solution was boiled until 50.0 cm^3 (50.0 g) of solution remained. The solubility of ethanedioic acid dihydrate, $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$ (M_r 126) is 14.3 g / 100 g of solution at 20°C .

Calculate the mass of ethanedioic acid dihydrate that crystallised from this solution at 20°C . [3]

Mass = g



- (c) The removal of alkyl fluorocarbons and CFCs from the environment presents a difficult problem.

One suggestion is to heat these halogenoalkanes with sodium ethanedioate. For example, difluoromethane reacts with sodium ethanedioate, $(\text{COO})_2\text{Na}_2$, to give hexafluorobenzene, sodium carbonate (Na_2CO_3) and water.

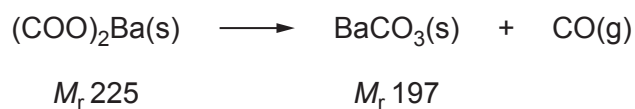
Write an equation for this reaction.

[2]

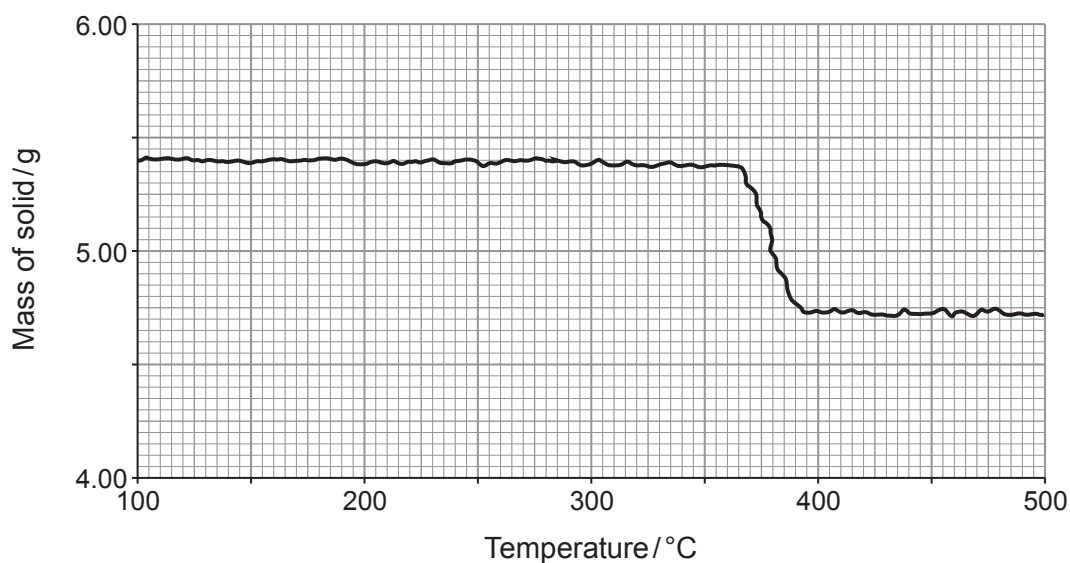
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- (d) On heating, barium ethanedioate undergoes decomposition.

Use the graph to show that the equation for this reaction is



[2]



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- (e) Decarboxylation occurs when carboxylic acids are heated with sodalime.

State the name of the organic compound produced when 4-ethylbenzenecarboxylic acid is heated with sodalime. [1]

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Examiner
only

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